Assignment 5 – Custom Marker Augmentation

Summary:

The objective of this assignment was to develop an Augmented Reality application that displays a 3D rotating Earth globe on top of a custom-designed image marker using the Vuforia Engine within Unity. The application uses marker-based tracking to detect the custom marker through a webcam or mobile camera and augment a textured 3D globe directly onto the marker surface in real time.

The Earth globe was created using a 3D sphere object with a realistic Earth texture applied to it. A continuous rotation script was added to simulate the rotation of the Earth. The globe was parented to the Image Target object so that the globe remains perfectly anchored to the marker during movement and tracking. The application successfully demonstrates real-time AR augmentation, marker tracking, and 3D object interaction using Unity and Vuforia technologies.

Technical Approach:

Unity was used as the development environment because it provides strong support for real-time 3D rendering, object management, lighting, scripting, and AR application development. The Vuforia Engine was used because it provides robust image target detection and marker tracking capabilities for marker-based AR applications.A custom image marker with high contrast and detailed features was designed and uploaded to the Vuforia Target Manager. Vuforia analyzed the marker and generated a tracking database containing .dat and .xml files for Unity integration.Inside Unity, the AR Camera was configured using the Vuforia license key. An Image Target object was added and linked to the uploaded marker database. The Earth globe was then added as a child object of the Image Target. This hierarchical structure ensured proper coordinate alignment between the physical marker and the virtual 3D globe. Since the globe inherits the position, rotation, and tracking information from the Image Target, the augmentation remains stable and correctly aligned with the marker in real time.

Vuforia Setup:

First, a Vuforia developer account was created using the official Vuforia Developer Portal. After logging in, a development license key was generated using the License Manager section. This license key was later used inside Unity to activate Vuforia services.Next, a custom image marker was designed with high contrast and unique visual features to improve tracking stability. The marker image was uploaded to the Vuforia Target Manager by creating a new database and adding a Single Image Target. Vuforia processed the image and assigned a feature rating based on marker quality. After successful processing, the database was downloaded as a Unity package containing the required .dat and .xml files.

Unity Integration:A new 3D Unity project was created and the Vuforia Engine package was imported into the project. The Main Camera was removed and replaced with the Vuforia AR

Camera. The generated Vuforia license key was pasted into the Vuforia configuration settings inside the AR Camera.The downloaded Vuforia database package was then imported into Unity. An Image Target object was added to the scene and linked to the uploaded marker database through the Inspector panel. A 3D sphere object was created to represent the Earth globe, and a realistic Earth texture was applied using a material.The Earth globe object was scaled and positioned appropriately, then parented under the Image Target object in the hierarchy. This ensured that the globe remained anchored to the marker during tracking. A C# rotation script was added to the globe to continuously rotate the Earth along its axis. Finally, the application was tested using a webcam, where the 3D globe successfully appeared and rotated on top of the custom marker in real time.